

Storage Management and Configuration

Disk drives have had partitions on them since the early IBM PC days. A partition is a division of a hard drive disk (HDD) into logically independent sections. The original requirement for a partition was to divide the disk into smaller chunks because the filesystem at the time could not consume the whole disk. The choices back then either used a 3rd party format tool or partitioned the disk into smaller pieces. It was common to use partitions, and the operating systems at that time supported up to four partitions. This partitioning scheme is known as MBR (Master Boot Record). Over time, the maximum of four partitions became a limiting factor; as such, extensions to the partitioning scheme were made: three primary partitions and one extended partition, where additional logical partitions could be created inside. All partitions greater than four are logical partitions (meaning contained within an extended partition). There can only be one extended partition, but it can be divided into numerous logical partitions. The maximum partitions per disk are still with us and documented in the *"Kernel Administration Guide"*.

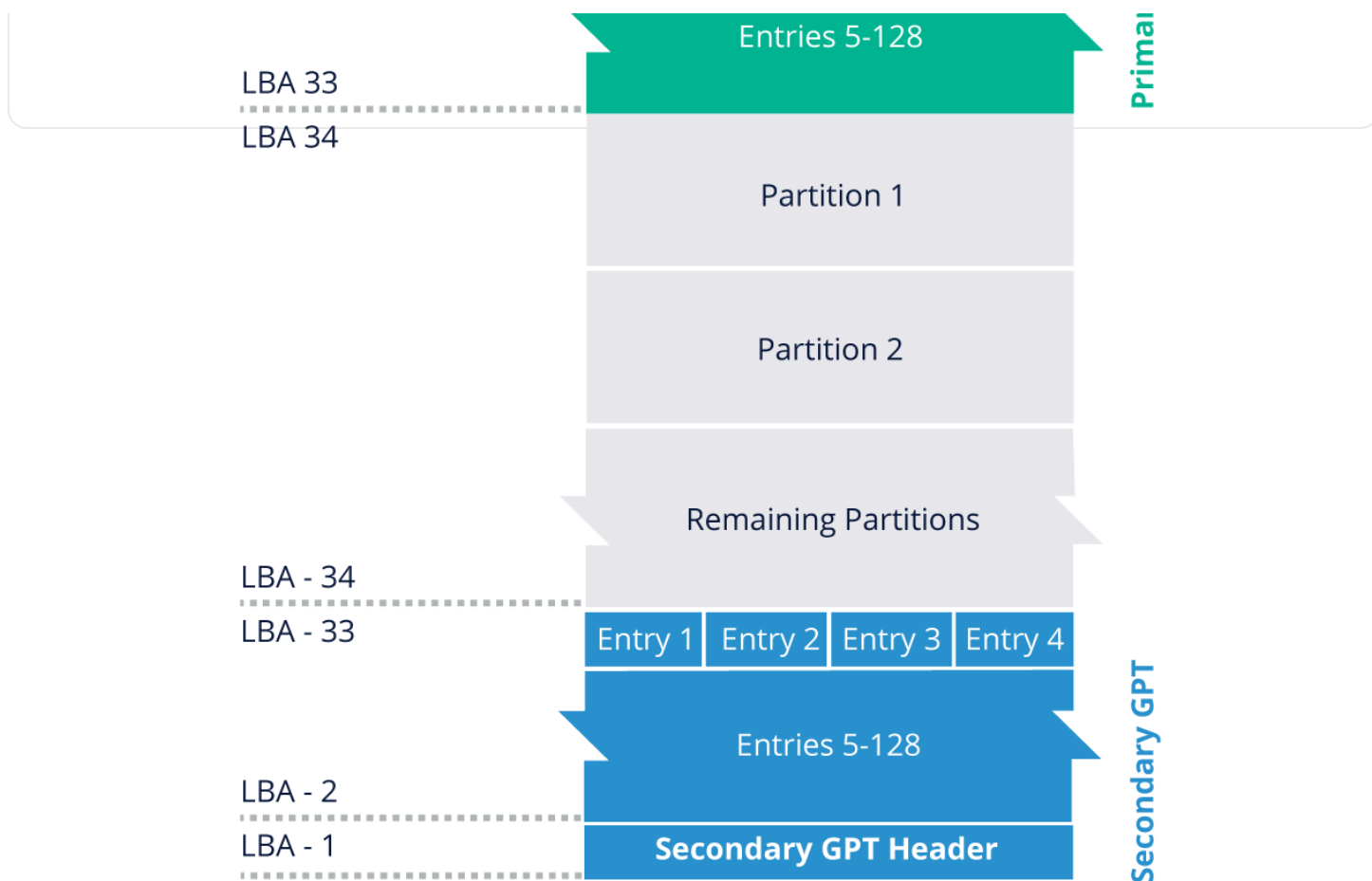
Table 5.3: MBR Scheme

PARTITION TYPE (PC-DOS)	DESCRIPTION
Primary	A primary partition can contain one filesystem or logical drive. Swap and boot are usually considered primary partitions.
Extended	An extended partition can contain several filesystems (logical drives) inside it. It has another partition table.
Logical	A logical partition is limited by the device type.

This MBR partitioning scheme still exists today. However, there is a *new* partitioning scheme called GPT (GUID Partition Table), where GUIDs are Globally Unique IDentifiers. The GPT can have up to 128 partitions per disk and is preferred in systems with a UEFI (Unified Extensible Firmware Interface), the replacement for PC-BIOS. Partitions help us section our storage drive into logically separate drives. This enables us to apply special parameters to sub-sections of the disk for security, performance, and other options.

GUID PARTITION TABLE SCHEME





Layout of a Disk With GPT.

Based on the [GUID Partition Table Scheme](#) image by Kbolino, from Wikipedia